

SOROUGH SABERAN

EMPLOYMENT HISTORY

SENIOR SOFTWARE ENGINEER

June 2024 - Present

Capital One

New York City

- Fully lead an ambitious initiative to build and deploy an AI monitoring service that would learn the early warning signs of incidents through unsupervised learning, and detect them in real time.
 - Designed, built, and trained an LSTM Autoencoder on OTel logs to learn normal behavior for an application, and to use that as the basis for predicting incidents, with great results. ~80% precision.
 - Also built the model to report explainability data, so that we know the reasons behind the predictions.
 - Built an electron app that utilized Claude to interpret the prediction and explainability results in a user-friendly way so that developers across Capital One could know the status of their apps. Also built into the electron app a way to use Claude to trigger human-in-the-loop agentic mitigation strategies when an incident would occur.
- Built the front end and middleware for a new internal Capital One website using React, JavaScript, and Typescript.
- Improved performance of a pre-existing machine learning-based incident prediction model by using XGBoost, improving its precision from ~25% to ~60%. Additionally integrated several new capabilities for incident prediction, including:
 - Incident severity prediction, leading to more effective risk mitigation.
 - SHAP to enable model explainability, identifying the causes of potential incidents.
 - Migrated predictions from running weekly to daily.

MACHINE LEARNING ENGINEER

March 2022 - June 2024

American Express

New York City

- Developed and implemented a suite of model interpretability and explainability tools, utilizing partial dependence plots, SHAP values, SHAP interaction values, and permutation feature importance.
- Engineered a proprietary gradient-boosted decision trees algorithm, similar to XGBoost, with enhanced features.
- Migrated several complex codebases from PySpark to Dask, achieving a 20% improvement in execution speed.

WEB DEVELOPER

August 2019 - March 2022

American Express

New York City

- Contributed to the development of US Personal Loans, US Business Loans, Amex Global Pay, and others.
- Reduced release cycle times by creating a full suite of automated browser tests, saving days of time testing per release.
- Proposed and implemented an initiative to ensure full compliance with Web Content Accessibility Guidelines (WCAG) across all web pages, delivering an accessible user experience.

Personal Project

JAMMYEGGS.APP

A web and mobile application for people to find and share content such as polls, quizzes, and much more.

- Sole creator. Leveraged Cursor and AI-assisted workflows to architect and deploy the full-stack application using React (JavaScript) and Supabase (TypeScript and PL/pgSQL).

SKILLS

AWS, Python, Scikit-Learn, XGBoost, Pandas, Dask, SQL, PL/pgSQL, PyTorch, JavaScript, TypeScript, React, Node.js, Express.js, FastAPI, WebdriverIO, Godot, Bash, Git, LaTeX, Jenkins, Databricks, Snowflake, Kubeflow, Supabase

EDUCATION

BACHELOR'S DEGREE IN MATHEMATICS-COMPUTER SCIENCE

March 2019

University of California-San Diego

San Diego, California

CERTIFICATION

AWS CERTIFIED SOLUTIONS ARCHITECT - ASSOCIATE

Expires April 27, 2028